

ADVANCED HTML & CSS – MASTER ASSIGNMENT || 11.59 PM || 27Th Feb 2026

SECTION A – ADVANCED HTML THEORY

1. Explain semantic vs non-semantic HTML elements with layout examples.

--> Semantic elements like header and main tell the browser what the content is. Non-semantic elements like div and span are just containers with no meaning.

2. Design a fully semantic blog page structure.

--> Use header for the title, nav for links, main for the content, article for posts, aside for the sidebar, and footer for the bottom.

3. Explain ARIA roles. When should ARIA be avoided?

--> ARIA roles like role="button" help screen readers. Avoid ARIA when a real HTML tag like button or nav already exists.

4. Create an accessible navigation structure using ARIA.

--> Use the nav tag with aria-label="Main Navigation" and use aria-current="page" on the link for the page you are on.

5. Explain aria-label, aria-labelledby, aria-describedby.

--> aria-label gives a name to an element with no text. aria-labelledby links to another element for a name. aria-describedby links to a hint or extra info.

6. Create a complex form using fieldset, legend, datalist, pattern validation.

--> Use fieldset to group inputs, legend for the group title, datalist for a search list, and pattern to check if the text is correct.

7. Explain structured data and schema markup.

--> It is hidden code that helps Google understand your content, like showing star ratings or prices in search results.

8. Create FAQ layout using details and summary.

--> The details tag creates a dropdown and the summary tag holds the question text.

9. Explain meta tags for SEO and performance.

--> Meta description helps search engines show a summary. Meta viewport helps the site fit correctly on mobile screens.

10. Build responsive images using picture, srcset, sizes.

--> srcset lets the browser pick the best image size for the user's screen width so the site loads faster.

11. Explain defer vs async.

--> Async runs the script as soon as it downloads. Defer waits until the HTML is fully finished before running the script.

12. Create accessible video embed with captions.

--> Use the video tag with controls and add a track tag inside it to provide subtitles for users.

13. Explain how HTML impacts performance metrics.

--> A clean and small HTML structure reduces DOM size, making the page load faster and improving performance scores.

14. Create multi-step form structure (HTML only).

--> Use multiple fieldset tags inside one form to separate different parts of the form logically.

15. Build semantic pricing table layout.

--> Use a section for each plan with a heading for the name, a price element, and a list for features.

SECTION B – ADVANCED CSS THEORY**16. Explain CSS specificity hierarchy.**

--> This is a score that decides which CSS rule wins. The order is: 1. Style inside the tag (Inline), 2. IDs (#name), 3. Classes (.name), 4. Tags (h1, p).

17. Explain stacking context.

--> This decides which items stay on top of others. It happens when you use z-index with position, or when you change opacity.

18. Compare all positioning types.

--> Static: Normal layout. Relative: Moves a bit from its home. Absolute: Moves inside its parent box. Fixed: Stays on screen when you scroll. Sticky: Stops at the top when you scroll past it.

19. Explain inheritance and cascade.

--> Inheritance: Children get styles like font and color from their parents automatically. Cascade: If two rules are the same, the one at the bottom of the file wins.

20. Deep explanation of Box Model.

--> Every element is a box. It has the Content (middle), Padding (inner space), Border (the line), and Margin (outer space).

21. Explain logical properties.

--> Instead of left or right, we use "start" and "end". This helps the layout flip correctly for languages that read right-to-left.

22. Compare em, rem, vh, vw, ch, clamp().

--> rem: Based on the main font size. vh/vw: Based on the screen height and width. clamp(): Sets a size that changes with the screen but stays between a min and max.

23. Explain CSS reset strategies.

--> Browsers have different default styles. A reset clears these out so your site looks the same in every browser.

24. Explain CSS containment.

--> This tells the browser that one part of the site is independent. It makes the page faster because the browser doesn't have to check everything when one part changes.

25. Explain CSS custom properties.

--> These are variables like --my-color. You can save a color once and use it everywhere, making it easy to change later.

26. Explain BEM methodology.

--> It is a way to name classes like .card__title--large. It keeps your code clean and prevents mistakes.

27. Explain prefers-reduced-motion.

--> A setting that checks if a user wants fewer animations. We use it to turn off moving effects for people who get dizzy easily.

28. Compare container queries and media queries.

--> Media Queries: Check the size of the whole screen. Container Queries: Check the size of the box the item is sitting in.

29. Explain will-change and isolation.

--> will-change: Tells the browser an animation is coming. isolation: Keeps an element's layers from getting mixed up with other elements.

30. Explain critical CSS.

--> The small bit of CSS needed for the very top of the page. Loading this first makes the site feel very fast to the user.

SECTION C – ADVANCED LAYOUT PRACTICAL

31. 12-column Grid System

--> A flexible layout divided into 12 equal parts that adjust for mobile and desktop screens.

32. Admin Dashboard Layout

--> A two-column setup with a fixed navigation sidebar on the left and a scrollable content area on the right.

33. Masonry Layout

--> A brick-style layout where items with different heights are packed tightly together to save space.

34. Hero Section with overlay

--> A large image or video at the top of a page with a semi-transparent dark layer to make text easier to read.

35. Sticky Sidebar Layout

--> A sidebar that stays visible in the viewport while the user scrolls through long content next to it.

36. Responsive Navbar with dropdown (CSS only)

--> A navigation bar that collapses on mobile and uses hover effects to show sub-menus without using JavaScript.

37. Animated Card Component

--> A content box that uses smooth transitions to move, shadow, or grow when the user interacts with it.

38. Pricing Section

--> A comparison layout showing different service tiers side-by-side using a flexible wrap system.

39. Scroll Snap Testimonial Slider

--> A horizontal list of reviews that automatically "snaps" to the center of each item during a swipe or scroll.

40. Multi-column Footer

--> An organized bottom section where links are grouped into several columns for better navigation.

41. CSS-only Accordion

--> A vertical list of headers that expand to show hidden text when clicked using only HTML tags.

42. Admin Panel Layout |

--> A professional layout featuring a top header bar for user settings and a side menu for site management.

43. Blog Grid Layout

--> A grid of article previews that automatically changes the number of columns based on the screen width.

44. CSS-only Modal

--> A popup window that appears over the main content by using unique ID links and CSS selectors.

45. Responsive Image Gallery

--> A grid of photos that keeps images aligned and perfectly sized across all devices.

SECTION D – ADVANCED EFFECTS & ANIMATIONS

46. Animated border button

--> A button where the border glows or moves in a circle when you hover over it.

47. CSS Loading Spinner

--> A simple circle that spins forever to show the user that the page is busy.

48. Text Reveal Animation

--> Words that slide up or fade in slowly when the page first loads.

49. CSS Parallax

--> An effect where the background image moves slower than the foreground text while scrolling.

50. Glassmorphism Card

--> A trendy design that looks like a sheet of blurry glass with a thin white border.

51. Neumorphism Component

--> A design style using soft shadows to make buttons look like they are pushed into or popping out of the background.

52. 3D Hover Card

--> A card that tilts or leans toward your mouse cursor when you hover over it.

53. Animated Gradient Background

--> A background where the colors slowly change or move from one side to the other.

54. CSS Tooltip

--> A small text box that pops up above a link or button to give extra information when you hover.

55. Skeleton Loading Animation

--> Gray boxes that pulse or shimmer to show where content will appear once it finishes loading.

SECTION E – MEDIA QUERIES & RESPONSIVE DESIGN

56. Implement 4 breakpoints (480px, 768px, 1024px, 1440px)

--> These are specific screen widths where the layout changes to fit mobile, tablet, laptop, and large desktop screens.

57. Convert layout to mobile-first

--> This means writing the simplest CSS for small screens first, then adding complexity for larger screens using min-width media queries.

58. Fluid typography using clamp()

--> A way to make text grow or shrink smoothly with the screen size so it is never too small on phones or too big on desktops.

59. CSS-only hamburger navigation

--> A mobile menu button created using a checkbox and a label that shows or hides links without using any JavaScript.

60. Orientation-based layout

--> Using CSS to change how the page looks specifically when a user turns their phone sideways (landscape) versus holding it upright (portrait).

61. Dark mode using prefers-color-scheme

--> A setting that automatically changes the website colors to dark if the user has "Dark Mode" turned on in their system settings.

62. Disable animations using prefers-reduced-motion

--> A safety setting that stops all moving or flashing effects for users who have motion sensitivity or get dizzy easily.

63. 5-breakpoint grid layout

--> A grid system that changes the number of columns five times (e.g., 1, 2, 3, 4, and 6 columns) as the screen gets wider.

64. Responsive pricing table

--> A list of price cards that stack vertically on small phones and sit in a horizontal row on wider computer screens.

65. Full responsive portfolio layout

--> A complete website design including a header, gallery, and contact section that adjusts perfectly to any device size.

SECTION F – FINAL ADVANCED PROJECTS

66. SaaS Landing Page

--> A professional page for software companies featuring a hero section, features list, and a call-to-action button.

67. eCommerce Product Page

--> A layout showing a product image, price, description, and an "Add to Cart" button that adjusts for mobile shoppers.

68. Streaming Platform Layout

--> A dark-themed design like Netflix with horizontal scrolling rows of movies and a large featured video at the top.

69. Multi-level Dropdown

--> A complex menu where hovering over one item opens a second menu, and hovering over that opens a third.

70. Responsive Dashboard

--> A screen for users to see data charts and stats that fits perfectly on both phones and wide monitors.

71. Blog Website Layout

--> A complete site with a main article list, category links, and a section for popular posts.

72. CSS Theme Switcher

--> A toggle button that lets users swap between different color themes (like Light, Dark, or Blue) using CSS variables.

73. Floating Label Form

--> A modern input form where the placeholder text moves up and shrinks when the user clicks inside to type.

74. Flexbox-only Landing Page

--> A full website layout built entirely using the Flexbox system to handle all rows and columns.

FINAL MASTER PROJECT – Complete Responsive Website

75. Complete Responsive Website

--> A final project combining everything: a sticky navbar, hero section, services grid, testimonials, and a contact footer.